MASTERARBEIT | MASTER'S THESIS

Titel | Title

Termination results for hybrid approach of Joint Spectral Radius computation

verfasst von | submitted by

Aaron Pumm, B.Sc.

angestrebter akademischer Grad | in partial fulfilment of the requirements for the degree of

Master Zusatz M.Sc.

Wien, | Vienna, 2026

Studienkennzahl lt. Studienblatt | degree programme code as it appears on the student record sheet:	UA 066 821
Studienrichtung lt. Studienblatt | degree programme as it appears on the student record sheet:	Masterstudium Mathematik
Betreut von | Supervisor:	Assoz. Prof. Dr. Martin Ehler
Ko-betreut von | Co-Supervisor:	Dr. Thomas Mejstrik MMag.

Contents

1 Introduction and Background **6**
1.1 Notation and Preliminaries . . . 7
1.2 Motivation and Applications of the Joint Spectral Radius . . . 7
1.3 Fundamental Theoretical Results . . . 9
1.4 Preprocessing and Normalization . . . 16

2 Invariant Polytope Algorithm **18**
2.1 Extremal and Invariant Norms . . . 18
2.2 Algorithmic Structure . . . 20
2.3 Stopping Criterion and Candidate Rejection . . . 20
2.4 Termination Conditions . . . 21
2.5 Rebalancing and Additional Initial Vertices . . . 21
2.6 Complex and Nonnegative Variants . . . 22

3 Finite-Tree Algorithm **23**
3.1 Tree-Based Notation and Definitions . . . 23
3.2 Algorithmic Structure . . . 25
3.3 Termination Conditions . . . 27

4 Hybrid Tree-Polytope Approach **28**
4.1 Hybridization Strategies . . . 28
4.2 Problem Formulation . . . 28
4.3 Hybrid Algorithm . . . 29
4.4 Termination Theory . . . 29
4.4.1 Proof Strategy . . . 29
4.4.2 Auxiliary Termination Results . . . 30
4.4.3 Main Termination Result . . . 32
4.5 Discussion . . . 32
4.5.1 Choice of Test Trees . . . 32
4.5.2 Eventual Polytope Invariance . . . 32
4.6 Implementation Considerations . . . 33

5 Numerical Experiments **34**
5.1 Runtime Analysis of Algorithm 3 on Generated Matrix Families . . . 34
5.2 Comparison with Algorithm 1 . . . 34
5.3 Comparison with Algorithm 2 . . . 34

5.4 Case Studies on Previously Open Problems 34

6 Conclusions and Outlook **35**
6.1 Summary of the Hybrid Framework 35
6.2 Resolved Problem Classes 35
6.3 Computational Efficiency 35
6.4 Remaining Open Problems 35
6.5 Directions for Future Research 35
6.5.1 Extension to the Complex Setting 35
6.5.2 Structured Nonnegative Matrix Families 35
6.5.3 Optimization of Tree Construction and Vertex Selection 35

Zusammenfassung

Diese Arbeit präsentiert einen hybriden Ansatz zur Berechnung des gemeinsamen Spektralradius (JSR) einer endlichen Menge von Matrizen. Die vorgeschlagene Methode integriert die Invariante-Polytop und Endlicher-Baum Algorithmen, um deren jeweilige Stärken zu nutzen. Die allgemeine Anwendbarkeit des baumbasierten Ansatzes wird mit der Effizienz der Polytop-basierten Techniken kombiniert. Der entwickelte Algorithmus konstruiert einen Baum-Graphen und überprüft maximale Produkte mithilfe angepasster Normen, wodurch eine Terminierung in Fällen ermöglicht wird, in denen klassische Methoden an ihre Grenzen stoßen, zum Beispiel wegen oberen Schranken bezüglich der Laufzeit. Theoretische Garantien werden bereitgestellt, um Korrektheit und Effizienz zu belegen. Es werden auch JSR-Berechnungen für Matrizenmengen, bei denen bestehende Methoden scheitern oder ineffizient sind, demonstriert.

Abstract

This thesis presents a hybrid approach for computing the joint spectral radius (JSR) of a finite set of matrices. The proposed method integrates the invariant polytope algorithm and the finite tree algorithm to leverage their respective strengths — integrating the vast applicability of the finite-tree-based approach with the efficiency of polytope-based techniques. The developed algorithm constructs a structured tree representation and verifies spectrum-maximizing products using adapted norms, enabling termination in cases where classical methods take too long or fail. Theoretical guarantees are provided to demonstrate correctness and efficiency.

Keywords: Joint spectral radius, hybrid algorithm, invariant polytope, finite tree method, matrix norms.

1 Introduction and Background

The *Joint Spectral Radius* (JSR) was first introduced by G.-C. Rota and G. Strang in 1960 (Rota and Strang, 1960). They described the JSR as the maximal exponential growth rate of a product of matrices taken from a finite set. Since its inception, the JSR has become a cornerstone in various mathematical and engineering disciplines due to its ability to encapsulate the asymptotic behavior of matrix products.

The concept gained significant traction in the 1990s when researchers began exploring its theoretical properties and practical implications. Notable advancements include its application in wavelet theory, where it assists in the construction of refinable functions (Daubechies and Lagarias, 1992), as well as in control theory, where it is used to analyze the stability of switched linear systems (Jungers, 2009), for which we provide a small example below. The computational challenges associated with determining the JSR have inspired the development of several algorithms, such as the invariant-polytope method (Guglielmi and Protasov, 2013) and the finite-tree method (Möller and Reif, 2014).

Despite the progress, the JSR computation remains a challenging problem, particularly due to the exponential complexity of exploring all possible matrix products. This thesis seeks to contribute to this ongoing effort by leveraging the invariant-polytope algorithm and the finite-tree algorithm to create a hybrid methodology that mitigates their respective limitations.

Thesis Outline

This thesis is structured as follows: Chapter 1 provides background on the JSR and its basic properties. Chapters 2 and 3 present the key ideas behind the algorithms used to construct the proposed hybrid approach, including their theoretical foundations and algorithmic implementations. Chapter 4 discusses possible combinations of the former approaches, proposes the so-called Tree-flavored invariant-polytope algorithm, and provides proofs of its termination properties, which form the main results of this thesis. Chapter 5 presents numerical results to analyze the efficiency and applicability of the hybrid algorithm. Chapter 6 concludes with insights and future directions.

1.1 Notation and Preliminaries

In this thesis, the spectral radius of a matrix A is denoted by $\rho(A)$. Unless stated otherwise, all matrices considered are assumed to have real entries, and all matrix norms are presumed to be submultiplicative. The analogous results for complex matrices follow similarly and may be reduced to the real case when necessary.

Let $\mathcal{A} = \{A_1, \ldots, A_n\}$ denote a finite collection of matrices, each of size $d \times d$.

For any positive integer $k \in \mathbb{N}$, we define the set of positive multi-indices of length k as

$$\mathcal{I}_k := \{1, \ldots, n\}^k.$$

Given a positive multi-index $I = [i_1, \ldots, i_k] \in \mathcal{I}_k$, we define the associated matrix product by

$$A_I = A_{[i_1,\ldots,i_k]} := A_{i_k} \cdots A_{i_1}.$$

Note that the indices are in reversed order compared to the multiplications.

The set of all possible products of length k formed from elements of $\mathcal{A}$ is then given by

$$\mathcal{A}^k := \{A_I \mid I \in \mathcal{I}_k\}.$$

For any scalar $\alpha \in \mathbb{R}$, we define the scaled matrix set by

$$\alpha\mathcal{A} := \{\alpha A_1, \ldots, \alpha A_n\}.$$

1.2 Motivation and Applications of the Joint Spectral Radius

Let $\mathcal{A} = \{A_1, \cdots, A_n\}$ be a finite set of $d \times d$ matrices. Consider a discrete-time linear switched system of the form:

$$x_{k+1} = A_{\sigma(k)} x_k, \quad x_0 \in \mathbb{R}^d, \tag{1.1}$$

where $\sigma : \mathbb{N} \to \{1, \ldots, n\}$ is a switching signal.

A key question is whether the system is *uniformly asymptotically stable*, i.e., whether every trajectory (x_k) converges to zero for all initial states and switching sequences:

$$\lim_{k\to\infty} x_k = 0.$$

This behavior is governed by a generalization of the spectral radius for a single matrix, the so-called *joint spectral radius (JSR)* of a matrix set $\mathcal{A}$ (Rota and Strang, 1960).

Definition 1.2.1. (Rota and Strang, 1960) For a finite set of matrices $\mathcal{A} = \{A_1, \ldots, A_n\}$, the JSR is defined as:

$$\mathrm{JSR}(\mathcal{A}) := \lim_{k\to\infty} \max_{P\in\mathcal{A}^k} \|P\|^{1/k}. \tag{1.2}$$

where $\|\cdot\|$ denotes any matrix norm.

This formulation captures the maximal asymptotic growth rate of matrix products.

Illustrative Example

Consider the matrix set:

$$\mathcal{A} = \left\{ A_1 = \begin{bmatrix} 0.5 & 1 \\ 0 & 0.5 \end{bmatrix}, \quad A_2 = \begin{bmatrix} 0.3 & 0 \\ 1 & 0.3 \end{bmatrix} \right\}.$$

Each matrix individually has spectral radius less than 1. However, their product

$$A_1 A_2 = \begin{bmatrix} 0.5 & 1 \\ 0 & 0.5 \end{bmatrix} \begin{bmatrix} 0.3 & 0 \\ 1 & 0.3 \end{bmatrix} = \begin{bmatrix} 1.3 & 0.3 \\ 0.15 & 0.15 \end{bmatrix}$$

has spectral radius of approximately

$$\rho(A_1 A_2) \approx 1.2929 > 1.$$

This example shows that switching can amplify the effects of individual matrices, leading to divergent behavior even if all single matrices are contractive. Thus, the JSR can exceed the spectral radius of any individual matrix in the set.

The following result motivates the importance of bounding or approximating $\mathrm{JSR}(\mathcal{A})$. It is known that:

- The system (1.1) is *uniformly bounded* (i.e., $\sup_k \|x_k\| < \infty$ for all $x_0 \in \mathbb{R}^d$) if and only if $\mathrm{JSR}(\mathcal{A}) \leq 1$ and $\mathcal{A}$ is irreducible.
- It is *uniformly asymptotically stable* if and only if $\mathrm{JSR}(\mathcal{A}) < 1$.

Understanding the joint spectral radius (JSR) is therefore critical for analyzing the asymptotic behavior of switched linear systems, as illustrated by the example above, and motivates the development of reliable algorithms for determining or approximating it (Jungers, 2009). Beyond control theory, the JSR plays a central role in several other areas. In approximation theory, it governs regularity questions for two-scale/refinement equations and wavelets (Daubechies and Lagarias, 1992; Maesumi, 2000). It is also used in the convergence and smoothness analysis of subdivision schemes (Guglielmi et al., 2011). In information theory, capacities of constrained codes can be studied through spectral characteristics of matrix families, leading to problems closely connected to JSR computation (Blondel et al., 2006). In discrete mathematics and combinatorics, JSR techniques appear in asymptotic counting problems such as partition-related functions and growth of structured matrix products (Protasov, 2000; Jungers et al., 2008, 2009). These connections highlight the JSR as a unifying concept across operator theory, approximation theory, dynamical systems, information theory, and discrete mathematics, and further motivate efficient computational methods such as the hybrid algorithm based on works of (Mejstrik and Reif, 2025), which will be explored in this thesis.

1.3 Fundamental Theoretical Results

Lemma 1.3.1 (Fekete's Lemma)**.** *Let $(a_n)_{n\geq 1}$ be a sequence of real, nonnegative numbers such that*

$$a_{m+n} \leq a_m + a_n \qquad \text{for all } m, n \geq 1. \tag{1.3}$$

Then the limit

$$\lim_{n\to\infty} \frac{a_n}{n}$$

exists and is equal to $\inf_{n\geq 1}\left(\frac{a_n}{n}\right)$.

Proof. Set $b_n := a_n/n$ and let $\alpha := \inf_{n\geq 1} b_n$. Fix $n \geq 1$ and write any $k \geq 1$ as $k = qn+r$ with integers $q \geq 0$ and $0 \leq r < n$. By subadditivity (1.3),

$$a_k = a_{qn+r} \leq qa_n + a_r.$$

Dividing by k gives

$$\frac{a_k}{k} \leq \frac{qa_n}{qn+r} + \frac{a_r}{qn+r} \leq \frac{a_n}{n} + \frac{a_r}{qn}.$$

Taking $\limsup_{k\to\infty}$ (equivalently $q \to \infty$) yields

$$\limsup_{k\to\infty} \frac{a_k}{k} \leq \frac{a_n}{n}.$$

Since n was arbitrary, we obtain

$$\limsup_{k\to\infty} \frac{a_k}{k} \leq \inf_{n\geq 1} \frac{a_n}{n} = \alpha.$$

On the other hand, by definition of infimum we always have $\alpha \leq a_k/k$ for all k, hence $\alpha \leq \liminf_{k\to\infty} a_k/k$. Therefore $\liminf = \limsup = \alpha$, so the limit exists and equals α. □

Theorem 1.3.2. *(Rota and Strang, 1960; Jungers, 2009) The JSR is well-defined and independent of the choice of the matrix norm.*

Proof. Let $\|\cdot\|_1$ and $\|\cdot\|_2$ be two matrix norms on $\mathbb{R}^{n\times n}$. By equivalence of norms in finite-dimensional vector spaces, there exist constants $c, C > 0$ such that:

$$c\|P\|_1 \leq \|P\|_2 \leq C\|P\|_1 \quad \forall P \in \mathbb{R}^{d\times d}.$$

Therefore, we have

$$\begin{aligned}
&\lim_{k\to\infty} \max_{P\in\mathcal{A}^k} \|P\|_1^{1/k} \leq \lim_{k\to\infty} \left(\frac{1}{c}\right)^{\frac{1}{k}} \max_{P\in\mathcal{A}^k} \|P\|_2^{1/k} \\
&= \lim_{k\to\infty} \max_{P\in\mathcal{A}^k} \|P\|_2^{1/k} \leq \lim_{k\to\infty} C^{\frac{1}{k}} \max_{P\in\mathcal{A}^k} \|P\|_1^{1/k} \\
&= \lim_{k\to\infty} \max_{P\in\mathcal{A}^k} \|P\|_1^{1/k},
\end{aligned}$$

which makes the limit independent of the choice of the used matrix norm. The existence of this limit is a direct consequence of Fekete's Lemma. □

To facilitate the analysis of the algorithms and presentation of the main results, we first establish fundamental properties of the JSR.

Homogeneity of the Joint Spectral Radius

Proposition 1.3.3. *(Jungers, 2009) The JSR is homogeneous, meaning for any set of matrices $\mathcal{A}$ and scalar $\alpha \in \mathbb{R}$ we have*

$$\mathrm{JSR}(\alpha\mathcal{A}) = |\alpha|\,\mathrm{JSR}(\mathcal{A}). \tag{1.4}$$

Proof. Let $\mathcal{A} = \{A_1, \dots, A_n\}$ and $\alpha \in \mathbb{R}$. Then:

$$\begin{aligned}
\mathrm{JSR}(\alpha\mathcal{A}) &= \lim_{k\to\infty} \max_{A_i \in \mathcal{A}^k} \|\alpha A_{i_k} \cdots \alpha A_{i_1}\|^{1/k} \\
&= |\alpha| \lim_{k\to\infty} \max_{A_i \in \mathcal{A}^k} \|A_{i_k} \cdots A_{i_1}\|^{1/k} \\
&= |\alpha| JSR(\mathcal{A})
\end{aligned}$$

□

Berger–Wang Characterization and Three-Member Inequality

Proposition 1.3.4. *(Berger and Wang, 1992) The* JSR *can equivalently be defined as*

$$\mathrm{JSR}(\mathcal{A}) = \limsup_{k\to\infty} \max_{P \in \mathcal{A}^k} \rho(P)^{1/k},$$

if $\mathcal{A}$ is a finite (or compact) set.

Proposition 1.3.5. *(Jungers, 2009) Let $\|\cdot\|$ be a matrix norm. Then for every $k \in \mathbb{N}$,*

$$\max_{P \in \mathcal{A}^k} \rho(P)^{1/k} \;\le\; \mathrm{JSR}(\mathcal{A}) \;\le\; \max_{P \in \mathcal{A}^k} \|P\|^{1/k}. \tag{1.5}$$

Proof. Fix $k \in \mathbb{N}$ and choose

$$P_{\max} \in \arg\max_{P \in \mathcal{A}^k} \rho(P)^{1/k}.$$

For the left-hand side we obtain, for any $r \in \mathbb{N}$,

$$\begin{aligned}
\max_{P \in \mathcal{A}^k} \rho(P)^{1/k} &= \rho(P_{\max})^{1/k} \\
&= \rho(P_{\max})^{r/(kr)} \\
&= \rho(P_{\max}^r)^{1/(kr)} \\
&\le \max_{P \in \mathcal{A}^{kr}} \rho(P)^{1/(kr)} \\
&\le \limsup_{r\to\infty} \max_{P \in \mathcal{A}^{kr}} \rho(P)^{1/(kr)} \\
&\le \limsup_{n\to\infty} \max_{P \in \mathcal{A}^n} \rho(P)^{1/n} \\
&= \mathrm{JSR}(\mathcal{A}).
\end{aligned} \tag{1.6}$$

For the right-hand side, set

$$M_n := \max_{P \in \mathcal{A}^n} \|P\| \qquad (n \in \mathbb{N}).$$

By submultiplicativity of the norm, every $Q \in \mathcal{A}^{m+n}$ can be written as $Q = RS$ with $R \in \mathcal{A}^m$ and $S \in \mathcal{A}^n$, hence

$$\|Q\| \leq \|R\| \, \|S\| \leq M_m M_n, \tag{1.7}$$

and taking maxima over $Q \in \mathcal{A}^{m+n}$ yields

$$M_{m+n} \leq M_m M_n \qquad \text{for all } m, n \in \mathbb{N}. \tag{1.8}$$

Define the real sequence $a_n := \log M_n$. Then from (1.8),

$$a_{m+n} = \log M_{m+n} \leq \log(M_m M_n) = \log M_m + \log M_n = a_m + a_n, \tag{1.9}$$

so $(a_n)_{n\geq 1}$ is subadditive. By Fekete's lemma, the limit $\lim_{n\to\infty} a_n/n$ exists and equals $\inf_{n\geq 1} a_n/n$. In particular, for the fixed k we have

$$\lim_{n\to\infty} \frac{a_n}{n} = \inf_{n\geq 1} \frac{a_n}{n} \leq \frac{a_k}{k}. \tag{1.10}$$

Exponentiating (1.10) gives

$$\begin{aligned} \lim_{n\to\infty} M_n^{1/n} &= \exp\left(\lim_{n\to\infty} \frac{a_n}{n} \right) \\ &\leq \exp\left(\frac{a_k}{k} \right) \\ &= \exp\left(\frac{\log M_k}{k} \right) \\ &= M_k^{1/k} = \max_{P\in\mathcal{A}^k} \|P\|^{1/k}. \end{aligned} \tag{1.11}$$

Since $\mathrm{JSR}(\mathcal{A}) = \lim_{n\to\infty} \max_{P\in\mathcal{A}^n} \|P\|^{1/n} = \lim_{n\to\infty} M_n^{1/n}$ by definition, the right-hand inequality in 1.5 follows. □

This inequality underlies many computational methods, since it yields computable lower and upper bounds on $\mathrm{JSR}(\mathcal{A})$. Moreover, as $k \to \infty$ the upper bound $\max_{P\in\mathcal{A}^k} \|P\|^{1/k}$ converges to $\mathrm{JSR}(\mathcal{A})$, while the lower bound $\max_{P\in\mathcal{A}^k} \rho(P)^{1/k}$ approaches $\mathrm{JSR}(\mathcal{A})$ in the sense of a $\limsup$.

Similarity Invariance and Reducibility

Proposition 1.3.6. *(Jungers, 2009) The* JSR *is invariant under similarity transformations, i.e., for any invertible matrix T we have*

$$\mathrm{JSR}(\mathcal{A}) = \mathrm{JSR}(T^{-1}\mathcal{A}T). \tag{1.12}$$

Proof. This follows from the definition of the JSR and the properties of the spectral radius.

$$\begin{aligned}
\mathrm{JSR}(T^{-1}\mathcal{A}T) &= \limsup_{k\to\infty} \max_{A_i\in\mathcal{A}^k} \rho(T^{-1}A_{i_k}T\cdot\ldots\cdot T^{-1}A_{i_1}T)^{1/k} \\
&= \limsup_{k\to\infty} \max_{A_i\in\mathcal{A}^k} \rho(T^{-1}A_{i_k}\cdot\ldots\cdot A_{i_1}T)^{1/k} \\
&= \limsup_{k\to\infty} \max_{A_i\in\mathcal{A}^k} \rho(A_{i_k}\cdot\ldots\cdot A_{i_1})^{1/k} \\
&= \mathrm{JSR}(\mathcal{A})
\end{aligned}$$

□

Definition 1.3.7. (Jungers, 2009) A set of matrices is called *(commonly) reducible* if there exists a nontrivial subspace of $\mathbb{R}^n$ that is invariant under all matrices in the set. This means there exists a change of basis that block-triangularizes all matrices in $\mathcal{A}$ at the same time. If $\mathcal{A}$ is not reducible, it is called *irreducible.*

Proposition 1.3.8. *(Protasov, 1997) If $\mathcal{A}$ is reducible, then*

$$\mathrm{JSR}(\mathcal{A}) = \max\{\mathrm{JSR}(\mathcal{B}), \mathrm{JSR}(\mathcal{D})\}, \tag{1.13}$$

where $\mathcal{B} = \{B_1, \ldots, B_n\}$ and $\mathcal{D} = \{D_1, \ldots, D_n\}$ are the blocks of the block-triangularized matrices

$$T^{-1}A_jT = \begin{bmatrix} B_j & C_j \\ 0 & D_j \end{bmatrix}$$

under some invertible matrix T.

Example 1.3.9.

$$A_1 = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}, \quad A_2 = \begin{bmatrix} 3 & 2 \\ 2 & 3 \end{bmatrix}$$

These matrices are reducible because there exists a nontrivial change of basis that transforms them into block-triangular form. The eigenvectors common to both matrices are

$$v_1 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}, \quad v_2 = \begin{bmatrix} 1 \\ -1 \end{bmatrix}.$$

Using the change of basis matrix

$$T = \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix},$$

we compute the similarity transformations

$$T^{-1}A_1T = \begin{bmatrix} 3 & 0 \\ 0 & 1 \end{bmatrix}, \quad T^{-1}A_2T = \begin{bmatrix} 5 & 0 \\ 0 & 1 \end{bmatrix}.$$

Since both matrices are upper triangular in this basis, they share a common invariant subspace, proving that they are reducible. Now, from Proposition 1.3.8, we know that the JSR can be calculated as

$$\mathrm{JSR}(\mathcal{A}) = \max\{\mathrm{JSR}(\{3,5\}), \mathrm{JSR}(\{1,1\})\} = 5$$

This can be applied iteratively until the sets of blocks are all irreducible. The problem is thereby split into similar problems of smaller dimension. For the following considerations we can now assume $\mathcal{A}$ to be irreducible.

Infimum Characterization by Norms

Proposition 1.3.10. *(Rota and Strang, 1960) The JSR can be equivalently defined as the infimum over all submultiplicative norms:*

$$\mathrm{JSR}(\mathcal{A}) = \inf_{\|\cdot\|} \max_{A\in\mathcal{A}} \|A\|. \tag{1.14}$$

Proof. Define the family of norms by

$$\|x\|_\epsilon := \max\left\{\left\|\frac{A}{\lambda+\epsilon}x\right\|_2 : A \in \mathcal{A}\right\},$$

where $\epsilon > 0$ and $\lambda := \mathrm{JSR}(\mathcal{A})$. Using the induced matrix norm, we obtain

$$\sup_{A\in\mathcal{A}} \|Ax\|_\epsilon \leq \lambda + \epsilon.$$

Since $\epsilon > 0$ is arbitrary, the infimum equals $\mathrm{JSR}(\mathcal{A})$. If $\mathcal{A}$ is irreducible, the infimum is attained (Berger and Wang, 1992). □

Finiteness Property and Spectrum-Maximizing Products

Definition 1.3.11. (Lagarias and Wang, 1995) A matrix set $\mathcal{A} = \{A_1, \ldots, A_n\}$ is said to possess the *finiteness property* if there exists a finite sequence of matrices $A_{i_1}, \ldots, A_{i_k}$ such that

$$\mathrm{JSR}(\mathcal{A}) = \|A_{i_k} \cdots A_{i_1}\|^{1/k}. \tag{1.15}$$

Any such product is referred to as a *spectrum-maximizing product* (s.m.p.).

Although the finiteness property does not hold for all matrix sets, it is crucial for the effectiveness and termination of most algorithmic approaches. This is shown, for example, in (Blondel et al., 2003).

Computational Complexity of JSR Computation

The computation of the joint spectral radius (JSR) is theoretically challenging due to several complexity results.

Approximating the JSR is **NP-hard** (Tsitsiklis and Blondel, 1997), meaning no polynomial-time algorithm is expected unless P = NP. Moreover, for general sets of matrices, even deciding whether the JSR is strictly less than one is **undecidable** (Blondel and Tsitsiklis, 2000). Despite these theoretical limitations, many practical cases allow for efficient numerical estimation.

These complexity results highlight fundamental challenges. There are still applied settings where efficient approximation algorithms provide useful results.

Candidate Products and Generators

Approximating the JSR, using the invariant-polytope or the finite-tree algorithms, requires identifying candidate products or *generators* of the matrix set that contribute most significantly to the asymptotic growth rate. These generators are often derived through optimization techniques and their identification is a key step in computational algorithms.

Spectral Gap

Definition 1.3.12. (Mejstrik and Reif, 2025, Definition 1.9) A finite matrix family $\mathcal{A}$ is said to possess a *spectral gap* if there exists a constant $\gamma \in [0, 1)$ such that for every $k \in \mathbb{N}$ and every product $A_I \in \mathcal{A}^k$,

$$\rho(A_I) = \mathrm{JSR}(\mathcal{A})^k \quad \text{or} \quad \rho(A_I) \leq \gamma \ \mathrm{JSR}(\mathcal{A})^k.$$

Equivalently, every non-s.m.p. product is uniformly separated from the maximal growth rate by the factor $\gamma < 1$.

Remark 1.3.13. (Mejstrik and Reif, 2025; Guglielmi and Protasov, 2024) In the invariant-polytope setting, a spectral gap is part of the standard finite-termination assumptions together with uniqueness (up to cyclic permutations and powers) and simplicity of the leading eigenvalue of the s.m.p. candidate. The hybrid/tree-based framework relaxes

this: once preprocessing yields a correctly scaled candidate with $\rho(\Pi) = 1$, proving existence of a suitable V-closed tree is sufficient to conclude $\mathrm{JSR}(\mathcal{A}) \leq 1$, hence no explicit spectral-gap estimate is required for correctness. We also note that (Guglielmi and Protasov, 2024) studies a closely related notion of gap as separation between the first and second Lyapunov exponents for matrix families.

Example 1.3.14. Consider

$$A_1 = \begin{bmatrix} 2 & 0 \\ 0 & 1 \end{bmatrix}, \qquad A_2 = \begin{bmatrix} 1 & 0 \\ 0 & 2 \end{bmatrix}, \qquad \mathcal{A} = \{A_1, A_2\}.$$

Since $\max\{\rho(A_1), \rho(A_2)\} = 2$ and $\|A_1\|_\infty = \|A_2\|_\infty = 2$, we get $\mathrm{JSR}(\mathcal{A}) = 2$. Any product $A_I \in \mathcal{A}^k$ has the form

$$A_I = A_1^m A_2^{k-m} = \begin{bmatrix} 2^m & 0 \\ 0 & 2^{k-m} \end{bmatrix},$$

so

$$\rho(A_I) = 2^{\max\{m, k-m\}}.$$

If $m = 0$ or $m = k$, then $\rho(A_I) = 2^k = \mathrm{JSR}(\mathcal{A})^k$ and A_I is an s.m.p. product. If $1 \leq m \leq k-1$, then

$$\rho(A_I) \leq 2^{k-1} = \tfrac{1}{2}\,\mathrm{JSR}(\mathcal{A})^k.$$

Hence Definition 1.3.12 holds with $\gamma = \frac{1}{2}$.

1.4 Preprocessing and Normalization

This thesis aims to address the challenge of computing the JSR by combining two existing algorithms that have demonstrated practical effectiveness in calculating the JSR for nontrivial sets of matrices. Both algorithms are based on the following simple concept:

We want to find the JSR of the finite set of matrices $\mathcal{A} = \{A_1, \cdots, A_n\}$.

1. **Assumptions**: $\mathcal{A}$ is irreducible.

2. **Candidates**: Efficiently find products $P = A_{i_k} \cdots A_{i_1}$ of matrices from $\mathcal{A}$ that maximize the averaged spectral radius $\hat{\rho} := \rho(P)^{\frac{1}{k}}$.

3. **Rescaling**: Transform $\mathcal{A} \to \tilde{\mathcal{A}}$ with $\tilde{A}_i := \frac{1}{\hat{\rho}} A_i$.

4. **Verification**: Establish that $\mathrm{JSR}(\tilde{\mathcal{A}}) = 1$ using the three-member inequality. By homogeneity, this is equivalent to $\mathrm{JSR}(\mathcal{A}) = \hat{\rho}$.

The considered algorithms differ only in step 4. The invariant-polytope algorithm tries to find a vector norm whose induced matrix norm already bounds the matrices from $\mathcal{A}$ sufficiently. The finite-tree algorithm, on the other hand, bounds products using a partitioning space that separates each product into terms bounded by 1 and a remainder term that does not grow fast enough to overcome the k-th root in the JSR definition (polynomial growth). By integrating these algorithms into a hybrid approach, this work advances the computational tools available for JSR analysis by combining efficiency with a broad class of matrix sets for which the algorithm terminates.

2 Invariant Polytope Algorithm

In this chapter, we focus on the invariant-polytope algorithm. At its core, this algorithm seeks a polytope $\mathcal{P} \subset \mathbb{R}^d$ that is invariant under the action of the already scaled matrices in $\tilde{\mathcal{A}}$. The corresponding Minkowski functional of this polytope is then a so-called extremal norm for the family $\mathcal{A}$. As we will see, this is enough to prove that the chosen candidate is indeed an s.m.p., which is our goal.

2.1 Extremal and Invariant Norms

Definition 2.1.1. (Barabanov, 1988) For a family of matrices $\mathcal{A}$ a norm $\|\cdot\|$ is called *invariant* or *Barabanov* if there exists a number $\lambda \geq 0$ such that

$$\max_{A \in \mathcal{A}} \|Ax\| = \lambda \|x\|, \quad \forall x \in \mathbb{R}^d.$$

It can be shown that such a λ will always equal the JSR.

Theorem 2.1.2. *(Barabanov, 1988) Every irreducible family $\mathcal{A}$ possesses an invariant norm.*

We are interested in norms with a weaker condition.

Definition 2.1.3. (Protasov, 1996) A norm $\|\cdot\|$ is called *extremal* for a family of matrices $\mathcal{A}$ if

$$\max_{A \in \mathcal{A}} \|Ax\| \leq \mathrm{JSR}(\mathcal{A}) \|x\| \quad \forall x \in \mathbb{R}^d.$$

Every invariant norm is, of course, extremal, which guarantees the existence of such a norm.

Theorem 2.1.4. *For every extremal norm $\|\cdot\|$ of a family of matrices $\mathcal{A}$*

$$\max_{P \in \mathcal{A}^k} \|P\| = \mathrm{JSR}(\mathcal{A})^k$$

holds for every $k \in \mathbb{N}$.

Proof. By submultiplicativity and extremality of the norm, we have

$$\begin{aligned}\max_{P\in\mathcal{A}^k}\|P\| &= \max_{d_k,\dots,d_1}\|A_{d_k}\cdots A_{d_1}\| \\ &\le \max_{d_k,\dots,d_1}\|A_{d_k}\|\cdots\|A_{d_1}\| \\ &\le \mathrm{JSR}(\mathcal{A})^k\end{aligned}$$

By the three-member inequality (1.5), we know

$$\begin{aligned}\mathrm{JSR}(\mathcal{A}) &\le \max_{P\in\mathcal{A}^k}\|P\|^{\frac{1}{k}} \\ \mathrm{JSR}(\mathcal{A})^k &\le \max_{P\in\mathcal{A}^k}\|P\|\end{aligned}$$

for every $k\in\mathbb{N}$.

□

Remark 2.1.5. In particular, when $k=1$, we have $\max_{A\in\mathcal{A}}\|A\| = \mathrm{JSR}(\mathcal{A})$. The existence of an extremal norm for $\mathcal{A}$ therefore enables the exact computation of the joint spectral radius.

In the following, we build an extremal norm whose unit ball is a balanced polytope, as introduced in (Protasov, 1996) and (Guglielmi and Zennaro, 2008).

Definition 2.1.6. A polytope $\mathcal{P}$ is called *balanced* if it spans the whole space and is equal to the symmetrized convex hull of a finite set of vertices.

$$\mathcal{P} = \mathrm{co_s}(V) := \mathrm{co}(V,-V) = \{x\in\mathbb{R}^d : x = \sum_{i=1}^{M}\alpha_i v_i \text{ and } \sum_{i=1}^{M}|\alpha_i|\le 1\}$$

If a polytope satisfies $\mathcal{A}\mathcal{P}\subseteq\mathcal{P}$ for some family $\mathcal{A}$, it is called *invariant under* $\mathcal{A}$.

Definition 2.1.7. (Magaril-Il'yaev and Tikhomirov, 2001) Let $C\subseteq\mathbb{R}^d$ be a nonempty, convex set such that $0\in\mathrm{int}(C)$. The *Minkowski functional* (or *gauge function*) associated to C is the mapping $p_C:\mathbb{R}^d\to[0,\infty)$ defined by:

$$p_C(x) = \inf\{\lambda>0 \mid x\in\lambda C\} \quad \forall x\in\mathbb{R}^d.$$

This is, in general, a seminorm on $\mathbb{R}^d$.

Remark 2.1.8. If $\mathcal{P}$ is a balanced polytope, then the associated Minkowski functional defines a norm on $\mathbb{R}^d$. Moreover, the corresponding induced matrix norm on $\mathbb{R}^{d\times d}$ is submultiplicative. In both cases, we denote the norm by $\|\cdot\|_{\mathcal{P}}$.

Theorem 2.1.9. *(Guglielmi and Protasov, 2013) If a balanced polytope $\mathcal{P}$ is invariant under $\tilde{\mathcal{A}}$ from Section 1.4, the corresponding Minkowski functional is an extremal norm for the family $\mathcal{A}$ and* $\mathrm{JSR}(\mathcal{A}) \leq \hat{\rho}$.

Proof.

$$\begin{aligned} \tilde{\mathcal{A}}\mathcal{P} \subseteq \mathcal{P} &\implies \mathcal{A}\mathcal{P} \subseteq \hat{\rho}\mathcal{P} \\ &\implies \|A\|_{\mathcal{P}} \leq \hat{\rho} \quad \forall A \in \mathcal{A} \\ &\implies \max_{A \in \mathcal{A}} \|A\|_{\mathcal{P}} \leq \hat{\rho} \end{aligned}$$

□

By the previous theorem, we only need to find a balanced polytope that is invariant under the action of $\tilde{\mathcal{A}}$. Then we have proved that the chosen candidate from the preprocessing is indeed an s.m.p.

2.2 Algorithmic Structure

After preprocessing has been carried out, we end up with a candidate $\tilde{\Pi} = \tilde{A_{d_k}} \cdots \tilde{A_{d_1}}$ for our finiteness-property hypothesis. We want to establish $\mathrm{JSR}(\tilde{\mathcal{A}}) = 1$ by finding an invariant balanced polytope $\mathcal{P} = \mathrm{co_s}(V)$. We define $V := \{v_1, \cdots, v_k\}$, where v_1 is the leading eigenvector of $\tilde{\Pi}$ (assumed to be real) and $v_i := \tilde{A_{d_{i-1}}} \cdot \ldots \cdot \tilde{A_{d_1}} v_1$ are the leading eigenvectors of the cyclic permutations of $\tilde{\Pi}$. Now we add new vertices to V iteratively by multiplying with matrices from $\tilde{\mathcal{A}}$, i.e.

$$V \overset{\cup}{\leftarrow} \mathcal{A}V :\iff V \leftarrow V \cup \mathcal{A}V.$$

All vertices that fall into the symmetrized convex hull of V can be disregarded, since they do not contribute to a change of the polytope $\mathcal{P}$ and therefore reduce computational effort. In practice, it is very important to drop as many vertices as possible, since V grows exponentially. This is done by solving a standard linear programming problem to determine whether $\|Av\|_{\mathrm{co_s}(V)} \geq 1$. If the algorithm does not produce new vertices that lie outside the current polytope for every $\tilde{A} \in \tilde{\mathcal{A}}$, then it is invariant by definition, and the finiteness property has been proved for the chosen candidate.

2.3 Stopping Criterion and Candidate Rejection

The runtime of Algorithm 1 can be very large. In general, it is at least exponential in the dimension. In cases where the s.m.p. candidate is not an s.m.p., or where there are multiple leading eigenvectors, the algorithm may fail to terminate. It is therefore

Algorithm 1 Invariant-polytope algorithm

```
V := {v_1, ··· , v_M}
V_new ← V
while V_new ≠ ∅ do
    V_rem ← V_new
    V_new ← ∅
    for v ∈ V_rem do
        for A ∈ 𝒜 do
            if ‖Av‖_{co_s(V)} ≥ 1 then
                V ⟵^∪ Av
                V_new ⟵^∪ Av
return co_s(V)
```

important to identify such cases early. To handle exponential growth, one sets an upper bound on the number of vertices in V. For rejecting an incorrect s.m.p. candidate, (Guglielmi and Protasov, 2013) proposed the following method. In the case where the candidate $\tilde{\Pi}$ has a unique simple leading eigenvalue and there exists only one s.m.p., define v_1^* as the leading eigenvector of $\tilde{\Pi}^*$ (the conjugate operator of $\tilde{\Pi}$), normalized by $(v_1, v_1^*) = 1$, and define $v_i^* := \tilde{A}_{d_i}^* \cdots \tilde{A}_{d_k}^* v_1^*$, which are the leading eigenvectors of the conjugates of the cyclic permutations of $\Pi \in \mathcal{A}^k$. If $\|Av\|_{\mathrm{co_s}(V)} > 1$ and there exists a j such that $|(v_j^*, Av)| > 1$, then the chosen candidate is not an s.m.p., and the algorithm either stops or restarts with a new candidate.

2.4 Termination Conditions

There are rare cases where the finiteness property does not hold, or where no invariant polytopes exist. In those cases, the algorithm will not find the value of the JSR.

2.5 Rebalancing and Additional Initial Vertices

Three years after publishing the fundamental algorithm with the stopping criterion, the authors released a new paper on rebalancing multiple s.m.p.s, as well as starting with extra vertices so that the polytope is better conditioned (Guglielmi and Protasov, 2016). In that setting, it is even possible to terminate with multiple s.m.p.s.

2.6 Complex and Nonnegative Variants

If the leading eigenvector associated with a candidate product is complex, then the real balanced polytope norm is no longer sufficient and must be replaced by an appropriate *complex-balanced polytope norm*. In addition, for families of nonnegative matrices one may exploit positivity to construct specialized norms that can substantially improve numerical efficiency. From an implementation perspective, these extensions typically affect only the membership test for the (symmetrized) convex hull, i.e., the underlying linear programming subproblem, while the overall structure of the algorithm remains unchanged. In the present thesis, we restrict attention to the real case in order to keep the exposition focused. For further details and algorithmic variants, we refer to (Guglielmi and Protasov, 2013; Guglielmi et al., 2005; Guglielmi and Zennaro, 2007).

3 Finite-Tree Algorithm

In this chapter we introduce basic definitions and concepts to another approach of JSR computation. It is called *finite-tree algrithm* as it is trying to decompose arbitrary products $P \in \mathcal{A}^k$ into sub-products that can be 1-bounded by an a priori chosen norm.

3.1 Tree-Based Notation and Definitions

Definition 3.1.1. A *tree* is a connected, acyclic graph T with a distinguished node t_0 called the *root*. The tree structure induces a natural hierarchy: for any node t_i, there exists a unique path from t_0 to t_i. The tree can then be described in terms of the following concepts:

- **Parent**: For any node $t \in T$, the node preceding t on the path from the root t_0 is called its *parent*.
- **Children**: The nodes adjacent to a given node t that are further from the root t_0 are called its *children*.
- **Leaf**: A node with no children is called a *leaf*.
- **Internal vertex**: A node with at least one child is called an *internal* or *non-leaf* node.

We define $T_{\mathcal{A}}$ as the tree consisting of all products in the set $\bigcup_{k \in \mathbb{N}_0} \mathcal{A}^k$ where every multi-index represents a node. For the empty index ($k = 0$) we set $A_{[]} = I_d$ the identity matrix in $\mathbb{R}^{d \times d}$. An example can be seen in Figure 3.1.

The objective is now to find a sub-tree $T \subset T_{\mathcal{A}}$ for a chosen norm, with the condition of every leaf encoding a product that has a norm less then 1 and every node that is not a leaf having all possible children $A_1, \cdots, A_J$. If such a tree has been found it easily allows for a decomposition of every possible product $P = A_{[i_1, \cdots, i_k]} \in \mathcal{A}^k$ into sub-products encoded by leafs and one possible rest term encoded by an internal-node. Leading to a proof of $\mathrm{JSR}(\tilde{\mathcal{A}}) = 1$.
Such sub-trees are in general not finite and therefore unusable for a computational tree-search algorithm. In the paper (Möller and Reif, 2014), the writers propose to use a different kind of encoding making use of so-called generators and set-valued nodes.

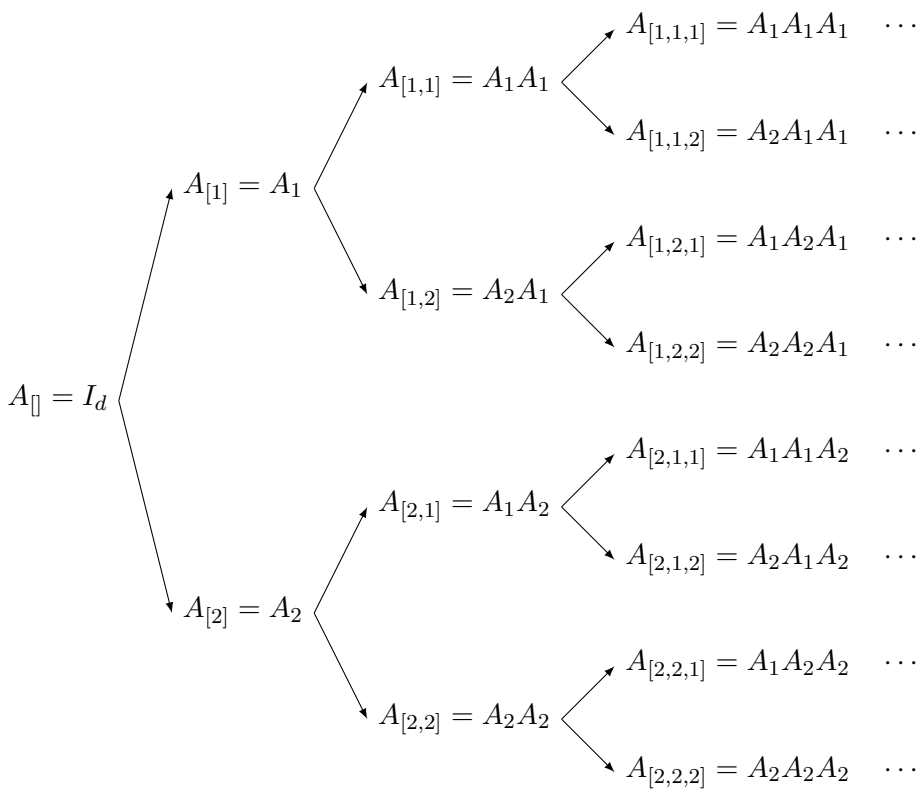

Figure 3.1: An example of the tree $T_{\mathcal{A}}$ on $\bigcup_{k\in\mathbb{N}_0} \mathcal{A}^k$ with $\mathcal{A} = \{A_1, A_2\}$.

We fix a set $\mathbf{G} = \{g_1, \cdots, g_m\}$ with $g_i \in \mathcal{A}_i^k$ and $\rho(g_i)^{\frac{1}{k_i}} \leq 1$. All products from this set will be called *generators*. We also define $A_l := g_{-l}$ for $l \in \{-m, \ldots, -1\}$.

Definition 3.1.2. (Möller and Reif, 2014)

The set $\mathcal{J}^k := \{-m, \ldots, -1, 1, \ldots, n\}^k$ denotes the collection of all multi-indeces of length k with entries in $\{-m, \ldots, -1, 1, \ldots, n\}$. For a multi-index $J = [j_1, \ldots, j_k] \in \mathcal{J}^k$, the corresponding matrix product set is defined by:

$$A_J := \{A_{j_k}^{e_k} \cdots A_{j_1}^{e_1} \mid e_i = 1 \text{ if } j_i > 0 \text{ or } e_i \in \mathbb{N}_0 \text{ otherwise}\}$$

Again, this creates a tree structure in a similar way to positive multi-indeces, except now the nodes represent (possibly infinite) sets of products.

Definition 3.1.3. Let $J = [j_1, \cdots, j_k]$ be a multi-index. We define the multi-index extension with $J + [j_{k+1}] := [j_1, \cdots, j_k, j_{k+1}]$.

your proposal $[J, j_{k+1}]$ is blunt since it would be nested like $[[j_1, \cdots, j_k], j_{k+1}]$

Definition 3.1.4. Let $t \in T$ be a node with the multi-index J. A child node $\tilde{t}$ with the multi-index $J + [j_{k+1}]$ is called:

- *positive* if j_{k+1} is positive.
- *negative* or *generator* if j_{k+1} is negative.

Now we want to find a subtree of a special form.

Definition 3.1.5. (Möller and Reif, 2014) A so-called $(\mathcal{A}, \mathbf{G})$-tree T has the following structure:

- The root node contains the identity matrix: $t_0 = \{I_d\}$.
- Each node $t \in T$ is either:
  - A leaf,
  - A parent of exactly n positive children.
  - A parent of only generators.

Definition 3.1.6. (Möller and Reif, 2014) A node in the tree is called *covered* if it is a subset of one of its ancestors in the tree. Otherwise, it is called *uncovered*. The set of uncovered leaves is denoted as

$$\mathcal{L}(T) := \{L \in t : t \in T \text{ is an uncovered leaf}\}.$$

and called *leafage* of the tree T.

Definition 3.1.7. (Möller and Reif, 2014) An $(\mathcal{A}, \mathbf{G})$-tree T is called *1-bounded* with respect to a matrix norm $\|\cdot\|$ if

$$\sup_{L \in \mathcal{L}(T)} \|L\| \leq 1.$$

If $\sup_{L \in \mathcal{L}(T)} \|L\| < 1$, then the tree is called *strictly 1-bounded*.

3.2 Algorithmic Structure

The finite-tree algorithm tries to find a 1-bounded $(\mathcal{A}, \mathbf{G})$-tree. We start with only the root node $t_0 = \{I\}$ and iteratively check in every step $\sup\{\|P\| : P \in L\} \leq 1$ for every leaf L in the current tree. If for one particular leaf the bound does not hold, we add either all positive children or an arbitrary amount of generator children.

Remark 3.2.1. (Möller and Reif, 2014) The procedure SELECTCHILDREN determines whether all positive children (from 1 to n) or a subset of negative children (from -1 to $-m$) is chosen, according to the theory established there.

Algorithm 2 Finite-tree-algorithm

Input: $\mathcal{A} = \{A_1, \ldots, A_n\}$ and $\mathbf{G} = \{g_1, \cdots, g_m\}$
Output: If the algorithm terminates the tree condition holds
$Q \leftarrow \{[1], \cdots, [n]\}$
while $Q \neq \emptyset$ **do**
  $Q_{\text{new}} \leftarrow \emptyset$
  for $J \in Q$ **do**
    if $\sup\{\|P\| : P \in A_J\} > 1$ **then**
      $\mathbf{J} \leftarrow \textsc{selectChildren}(J, \mathcal{A}, \mathbf{G})$
      $Q_{\text{new}} \overset{\cup}{\leftarrow} \{J + j : j \in \mathbf{J}\}$
  $Q \leftarrow Q_{\text{new}}$

Lemma 3.2.2. *For every $(\mathcal{A}, \mathbf{G})$-tree T there exists a strictly-positive, monotone polynomial p such that for every node $t \in T$ and every product $P \in \mathcal{A}^k$ encoded by t, $\|P\| \leq p(k)$ holds.*

Proof. Take an arbitrary node $t \in T$ with multi-index $J = [j_1, \cdots, j_l]$. Now every j_i encodes either a matrix from $\mathcal{A}$ or arbitrary powers of a generator from $\mathcal{G}$. Since all those matrices have spectral radius less then or equal to 1, its Jordan normal forms grow utmost polynomially und thus, due to equivalence of norms in finite-dimensional vector spaces, there exists a strictly-positive, monotone polynomial p_i with

$$\|A_{j_i}^e\| \leq p_i(e) \quad \forall e \in \mathbb{N}_0.$$

We define:

$$p_t := p_l \cdots p_1$$
$$p := \sum_{t \in T} p_t$$

The sum over all of T is finite since T has only finitely many nodes. Take now an arbitrary product $P = A_{j_l}^{e_l} \cdots A_{j_1}^{e_1} \in \mathcal{A}^k$ that is encoded by t.

Now we have:

$$\begin{aligned}
\|P\| &= \|A_{j_l}^{e_l} \cdots A_{j_1}^{e_1}\| \\
&\leq \|A_{j_l}^{e_l}\| \cdots \|A_{j_1}^{e_1}\| \\
&\leq p_l(e_l) \cdots p_1(e_1) \\
&\leq p_l(k) \cdots p_1(k) \\
&= p_t(k) \leq p(k)
\end{aligned}$$

Since the node t and the product P were chosen arbitrary, this concludes the proof. □

Theorem 3.2.3. *(Möller and Reif, 2014) If a 1-bounded $(\mathcal{A}, \mathbf{G})$-tree was found for a given matrix set $\mathcal{A}$ and generator set $\mathbf{G}$ then* $\mathrm{JSR}(\mathcal{A}) = 1$

Proof. We take an arbitrary positive multi-index $I = [i_1, \cdots, i_k]$. Our found $(\mathcal{A}, \mathbf{G})$-tree T allows for a unique decomposition of this multi-index into $I_k, \cdots, I_1$ where

$$A_I = A_{I_k} A_{I_{k-1}} \cdots A_{I_1}.$$

Each sub-multi-index $I_j \quad j = 1, \cdots, k-1$ encodes either a leaf of T or is the empty set and I_k encodes any node of T. Now due to Lemma 3.2.2 there exists a polynomial p such that

$$\|A_I\| = \|A_{I_k} A_{I_{k-1}} \cdots A_{I_1}\| \leq \|A_{I_k}\| \cdot \|A_{I_{k-1}}\| \cdots \|A_{I_1}\| \leq \|A_{I_k}\| \leq p(k).$$

The last equality holds since all $I_j \quad j = 1, \cdots, k-1$ are either the empty multi-index or leafs such that their products A_{I_j} are bounded by 1. Furthermore A_{I_k} can be bounded by the polynomial p and the length of the product, which is less then k. Now we have

$$\mathrm{JSR}(\mathcal{A}) = \lim_k \max_{P \in \mathcal{A}^k} \|P\|^{\frac{1}{k}} \leq \lim_k p(k)^{\frac{1}{k}} = 1$$

since the chosen multi-index was arbitrary. □

Remark 3.2.4. (Möller and Reif, 2014) The decomposition used in the proof of Theorem 3.2.3 exists and is unique due to the structure of an $(\mathcal{A}, \mathbf{G})$-tree. glueing tree on covered nodes creating an infinite tree that allows huffman encoding directly. Empty sets for every I_i if product already lies on internal node before I_k

3.3 Termination Conditions

Remark 3.3.1. (Möller and Reif, 2014) In the cited work, the authors show that the use of generators is essential for ensuring termination in cases where the original invariant-polytope algorithm fails to terminate. But deciding when to use generators instead of all positive children is still part of active research and is handled via highly heuristical approaches.

4 Hybrid Tree-Polytope Approach

In this chapter we will explore some possible combinations of the aforementioned algorithms and then present the main result of this work. The tree-flavored-invariant-polytope-algorithm and its termination results.

4.1 Hybridization Strategies

In its heart the invariant-polytope algorithm tries to find a polytope, whose corresponding Minkowski norm is specifically optimized on the given problem, whilst the finite-tree algorithm connects growth-rate to decompositions of products. A clear combination scheme arises naturally, where we use the optimized polytope norm to estimate the products of the finite-tree. From there we can choose a specific order or level of concurrency.
A modular approach would be to first run the invariant-polytope algorithm for a couple of steps and then use the calculated norm thats already optimized for the finite-tree algorithm. But that seems to be wasteful since valuable matrix calculations from the finite-tree algorithm could have been used for an even more optimized norm and some polytopes might have already cleared insight for the decompositions that the finite-tree algorithm tries to find. In (Mejstrik and Reif, 2025) the authors came up with a more concurrent algorithm that builds up norms and decompositions in every step.

4.2 Problem Formulation

After the preprocessing stage from Section 1.4 is done, we end up with a scaled set of matrices $\tilde{\mathcal{A}}$ and from the chosen candidate and the three-member-inequality (1.5) we already know $\mathrm{JSR}(\tilde{\mathcal{A}}) \geq 1$. Such that our goal of establishing $\mathrm{JSR}(\tilde{\mathcal{A}}) = 1$ becomes equivalent to $\mathrm{JSR}(\tilde{\mathcal{A}}) \leq 1$. For this we want to show that there exists a polynomial p and a norm $\|\cdot\|$, such that, similarly to Chapter 3, every norm of a product of matrices from $\tilde{\mathcal{A}}$ can be bounded by this polynomial in its length i.e.

$$\|P\| \leq p(k) \quad \forall P \in \tilde{\mathcal{A}}^k, k \in \mathbb{N}.$$

We find the polynomial p by decomposing every multi-index $J = [j_1, ..., j_n]$ into a finite series of 1-bounded sub-products followed by one leaf node $J = J_1 + ... + J_r + J_{leaf}$. We

dont know the leaf-subproduct for every multi-index directly but we can bound it by the theory of Chapter 3. The following describes the process of the developed algorithm together with its pseudocode, states the main theorem, outlines the proof and finally gives all theoretical results to prove this statement. We want to conclude this Chapter with some theoretical remarks as well as technical details for implementation.

4.3 Hybrid Algorithm

The algorithm starts just like the invariant-polytope algorithm with an set V_0 of starting-vertices. Every set of Vertices V_k defines a vector-polytope-norm $\|\cdot\|_{V_k}$. Now, at step k for every vertex $v \in V_k$ we take a test-$(\mathcal{A}, \mathbf{G})$-tree T_v. If there exists a matrix L in the leafage of T_v with $\|Lv\|_{V_k} \geq 1$ we add Lv to the set V_{k+1}. If all test-tree-leafages within a step k are 1-bounded by the polytope-norm, the algorithm terminates.

Algorithm 3 Tree-flavored-invariant-polytope-algorithm

Input: $\mathcal{A} = \{A_1, \ldots, A_n\}$, $\mathbf{G} = \{g_1, \cdots, g_m\}$ and $V = \{v_1, \cdots, v_M\}$
Output: A boolean value indicating whether the tree condition holds (True/False)
$V_{\text{new}} \leftarrow V$
while $V_{\text{new}} \neq \emptyset$ **do**
  $V_{\text{rem}} \leftarrow V_{\text{new}}$
  $V_{\text{new}} \leftarrow \emptyset$
  for $v \in V_{\text{rem}}$ **do**
    Construct$(\mathcal{A}, \mathbf{G})$-tree $\mathbf{T}$ with sepcifications from (Mejstrik and Reif, 2025)
    if $\exists L \in \mathcal{L}(T)$ with $\|Lv\|_{\text{cos}(V)} \geq 1$ **then**
      $V \overset{\cup}{\leftarrow} \mathcal{A}v$
      $V_{\text{new}} \overset{\cup}{\leftarrow} \mathcal{A}v$
return True

4.4 Termination Theory

Theorem 4.4.1. *If algorithm 3 terminates* $\text{JSR}(\tilde{\mathcal{A}}) \leq 1$.

4.4.1 Proof Strategy

Figure 4.1 shows an example of a generated tree after the algorithm has terminated as well as the $(\mathcal{A}, \mathbf{G})$-trees that have been used to bound the products. Here $\mathcal{A}$ consists of

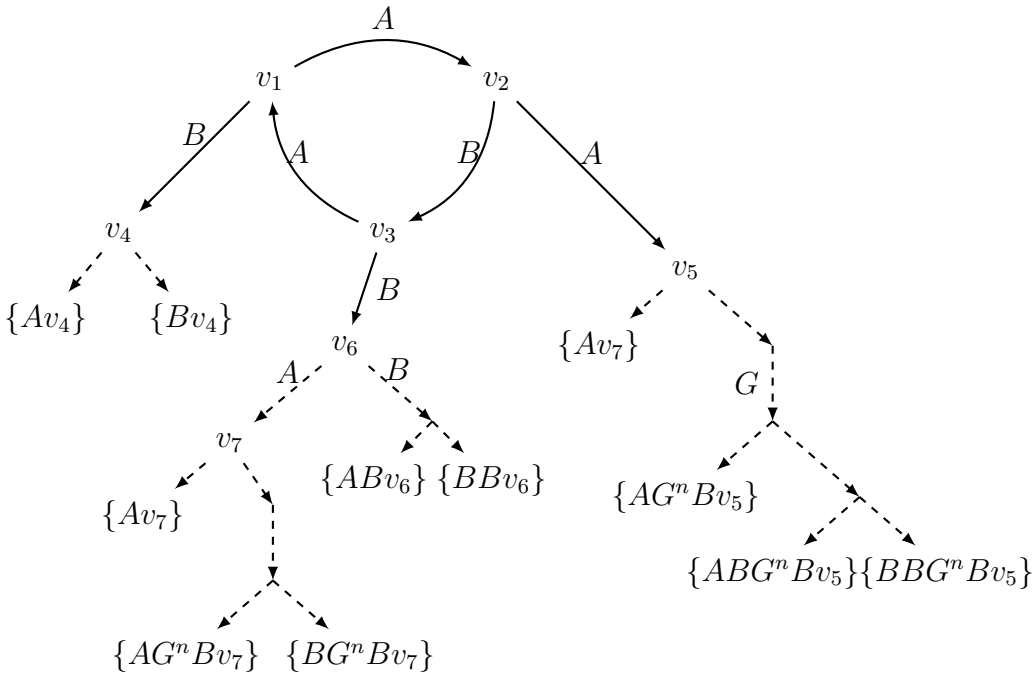

Figure 4.1: Cyclic tree structure generated by the algorithm. Vertices added to V (solid arrows) and finite-trees for bounding products (dashed arrows).

two matrices A and B and the chosen candidate is $\Pi = ABA$. At the top, you can see the eigenvector v_1 and the rest of the cycle, generated by the candidate Π, as well as branches coming of. All the nodes that are connected through a solid arrow have been added to the vertices V in a particular step k. All dashed lines show a test tree that is 1-bounded. You can see the 1-bounded matrix-vector products at the end of the dashed arrows where n can be every natural number including 0. Note that these products are bounded by the norm constructed in their respective steps but due to submultiplicativity the final norm bounds all products in general. So in the following we just reference the final norm $\|\cdot\|_V$ for the final set of vertices V.

explanations here: using testtrees, linear-combination trick for recursive bounding, huffmancoding, glueing from covered nodes into infinite tree, polynomial, growth rates, starting vertices, flat polytopes, etc.

4.4.2 Auxiliary Termination Results

Lemma 4.4.2. *If algorithm 3 terminates there exists an monotone polynomial p such that for every product $P \in \mathcal{A}^k$ and $v \in V$, $\|Pv\|_{cos(V)} \leq p(k)$ holds.*

Proof. If the algorithm terminates, then the set of vertices $V = \{v_1, \cdots, v_m\}$ is finite and for each $v \in V$ there exists an $(\mathcal{A}, \mathbf{G})$-tree T_v such that $\|Lv\|_{\cos(V)} \leq 1$ for every $L \in \mathcal{L}(T_v)$, which means there exists $\lambda_1, \cdots, \lambda_m$ such that $Lv = \sum \lambda_j v_j$ with $\sum |\lambda_j| \leq 1$.

According to Lemma 3.2.2, there exists a polynomial p_v such that for every product $P \in \mathcal{A}^k$ encoded by T_v, $\|Pv\|_{\mathrm{cos}(V)} \leq p_v(k)$. Take an arbitrary product $P \in \mathcal{A}^k$ and $v \in V$. Define:

$$\begin{aligned}
p &:= \sum_{v \in V} p_v \\
P'L &:= P \text{ with } L \in \mathcal{L}(T_v) \text{ if } P \notin T_v \text{ and } L := I \text{ else} \\
P'_{i_1 \dots i_r} L_{i_1 \dots i_r} &:= P'_{i_1 \dots i_{r-1}} \text{ with } L_{i_1 \dots i_r} \in \mathcal{L}(T_{v_{i_r}}) \text{ if } P'_{i_1 \dots i_{r-1}} \notin T_{v_{i_r}} \\
\text{and } L_{i_1 \dots i_r} &:= I \text{ else.} \\
\text{Where } i_j \in \mathcal{I} &:= \{1, \cdots, m\}
\end{aligned}$$

These decompositions exist and are unique because of the special structure of $(\mathcal{A}, \mathbf{G})$-trees. A simple prrof can be found in the appendix [appendix proof of decomp] and an in-depth explanaition of finding the decomposition can be found in (Möller and Reif, 2014).
Now we have:

$$\begin{aligned}
\|Pv\|_{cos(V)} &= \|P'Lv\|_{cos(V)} \\
&= \|P' \sum_{i_1 \in \mathcal{I}} \lambda_{i_1} v_{i_1}\|_{cos(V)} \\
&= \sum_{i_1 \in \mathcal{I}} |\lambda_{i_1}| \|P' v_{i_1}\|_{cos(V)} \\
&= \sum_{i_1 \in \mathcal{I}} |\lambda_{i_1}| \|P'_{i_1} L_{i_1} v_{i_1}\|_{cos(V)} \\
&= \sum_{i \in \mathcal{I}^2} |\lambda_{i_1}| |\lambda_{i_2}| \|P'_i v_{i_2}\|_{cos(V)} \\
&\dots \\
&= \sum_{i \in \mathcal{I}^k} |\lambda_{i_1}| \cdot \dots \cdot |\lambda_{i_k}| \|P'_i v_{i_k}\|_{cos(V)} \\
&\leq \sum_{i \in \mathcal{I}^k} |\lambda_{i_1}| \cdot \dots \cdot |\lambda_{i_k}| \; p_{v_{i_k}}(k) \\
&\leq p(k) \sum_{i \in \mathcal{I}^k} |\lambda_{i_1}| \cdot \dots \cdot |\lambda_{i_k}| \\
&= p(k) \sum_{i_1 \in \mathcal{I}} |\lambda_{i_1}| \sum_{i_2 \in \mathcal{I}} |\lambda_{i_2}| \dots \sum_{i_k \in \mathcal{I}} |\lambda_{i_k}| \\
&\leq p(k) \cdot 1 \cdot \dots \cdot 1 \\
&\leq p(k)
\end{aligned}$$

Here the first inequality works since after k steps the prefix P'_i is guaranteed to be in the tree $T_{v_{i_k}}$ and thus the norm is bounded by $p_{v_{i_k}}(k)$, which in turn is of course less then $p(k)$ by definition. The second to last inequality works since those sums of the absolute values of the coefficients of a linear combination of vectors are less than or equal to 1 from the 1-boundedness of the $(\mathcal{A}, \mathbf{G})$-trees. □

4.4.3 Main Termination Result

Proof. Due to the prework in the lemmas theorem 4.4.1 is now easy to show. Lemma 4.4.2 implies that for every $P \in \mathcal{A}^k$ and $v \in V$, $\|Pv\|_{\mathrm{cos}(V)} \leq p(k)$ trivially this implies:

$$\forall P \in \mathcal{A}^k : \|P\|_{\mathrm{cos}(V)} \leq p(k)$$

Now we take the definition of the JSR and get:

$$\mathrm{JSR}(\mathcal{A}) = \lim_{k \geq 1} \max_{P \in \mathcal{A}^k} \|P\|^{\frac{1}{k}} \leq \lim_{k \geq 1} p(k)^{\frac{1}{k}} = 1$$

□

Of course, if the candidate had a spectral radius of 1 $\implies$ $\mathrm{JSR}(\mathcal{A}) = 1$.

Corollary 4.4.3. *If the chosen trees are always $T_v = \mathcal{A}$, then we would have exactly the invariant-polytope algorithm 1. Which makes the new approach a real generalization of the original.*

4.5 Discussion

4.5.1 Choice of Test Trees

It was proven in (Möller and Reif, 2014) that in order to have a bigger solution space then algorithm 1 the use of generators is mandatory. The choice of the testing-trees is not trivial and subject to active research. We propose to use so-called *minimal trees* that fulfill every necessary and beneficial condition of the $(\mathcal{A}, \mathbf{G})$-trees but keep the amount of nodes minimal.

4.5.2 Eventual Polytope Invariance

Since the polytope-norms of matrices in $\mathcal{A}$ are allowed to be bigger than 1 just less than $p(1)$ it is possible and very likely the construccted polytope is not invariant under $\mathcal{A}$ in general anymore but eventually invariant.

4.6 Implementation Considerations

which trees are used, how does the bounding of leafage work, approximations if over max iterations, differences to before, memory layout, exponential growth exponent/branching, memory and speed in Landau, ideas in parallelism

Definition 4.6.1. An $(\mathcal{A}, \mathbf{G})$-tree with just one generator at the root, one path from root to the covered node and all neccessarily added branches, to make it comply with definition 3.1.5, is called a *minimal tree*. This tree only depends on the set $\mathcal{A}$ and the chosen generator G. Its denoted by $T_{(\mathcal{A},\mathbf{G})}$ and if the context of $\mathcal{A}$ is clear, it is shortened to $T_{\mathbf{G}}$.

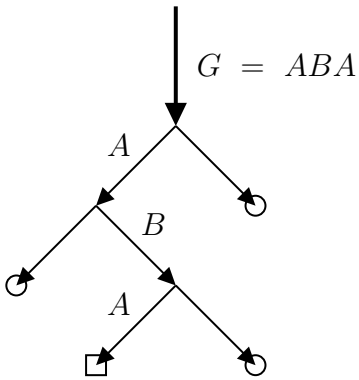

Figure 4.2: Minimal tree structure $T_{(G,\mathcal{A})}$ for the candidate $G = ABA$ and $\mathcal{A} = \{A, B\}$. The arrows marked with circles are uncovered-leafs, the arrow marked with a square is a covered-leaf

5 Numerical Experiments

5.1 Runtime Analysis of Algorithm 3 on Generated Matrix Families

5.2 Comparison with Algorithm 1

5.3 Comparison with Algorithm 2

5.4 Case Studies on Previously Open Problems

6 Conclusions and Outlook

6.1 Summary of the Hybrid Framework

6.2 Resolved Problem Classes

6.3 Computational Efficiency

6.4 Remaining Open Problems

6.5 Directions for Future Research

6.5.1 Extension to the Complex Setting

6.5.2 Structured Nonnegative Matrix Families

6.5.3 Optimization of Tree Construction and Vertex Selection

Bibliography

Alizadeh, F. and Goldfarb, D. (2003). Second-order cone programming. *Mathematical Programming*, 95(1):3–51.

Andersen, E. D., Roos, C., and Terlaky, T. (2003). On implementing a primal-dual interior-point method for conic quadratic optimization. *Mathematical Programming*, 95(2):249–277.

Ando, T. and Shih, M.-H. (1998). Simultaneous contractibility. *SIAM Journal on Matrix Analysis and Applications*, 19(2):487–498.

Barabanov, N. E. (1988). Lyapunov indicator for discrete inclusions, i–iii. *Automation and Remote Control*, 49(2):152–157.

Berger, M. A. and Wang, Y. (1992). Bounded semigroups of matrices. *Linear Algebra and its Applications*, 166:21–27.

Berstel, J. (2005). Growth of repetition-free words—a review. *Theoretical Computer Science*, 340(2):280–290.

Blondel, V. D., Jungers, R. M., and Protasov, V. Y. (2006). On the complexity of computing the capacity of codes that avoid forbidden difference patterns. *IEEE Transactions on Information Theory*, 52(11):5122–5127.

Blondel, V. D. and Nesterov, Y. (2005). Computationally efficient approximations of the joint spectral radius. *SIAM Journal on Matrix Analysis and Applications*, 27(1):256–272.

Blondel, V. D., Nesterov, Y., and Theys, J. (2005). On the accuracy of the ellipsoid norm approximation of the joint spectral radius. *Linear Algebra and its Applications*, 394:91–107.

Blondel, V. D., Theys, J., and Vladimirov, A. A. (2003). An elementary counterexample to the finiteness conjecture. *SIAM Journal on Matrix Analysis and Applications*, 24(4):963–970.

Blondel, V. D. and Tsitsiklis, J. N. (2000). Approximating the spectral radius of sets of matrices in the max-algebra is NP-hard. *IEEE Transactions on Automatic Control*, 45(9):1762–1765.

Cassaigne, J. (1993). Counting overlap-free binary words. In *STACS 93*, volume 665 of *Lecture Notes in Computer Science*, pages 216–225, Berlin. Springer.

Cicone, A., Guglielmi, N., Serra-Capizzano, S., and Zennaro, M. (2010). Finiteness property of pairs of 2
*times*2 sign-matrices via real extremal polytope norms. *Linear Algebra and its Applications*, 432(2–3):796–816.

Daubechies, I. and Lagarias, J. C. (1992). Two-scale difference equations. II. local regularity, infinite products of matrices and fractals. *SIAM Journal on Mathematical Analysis*, 23(4):1031–1079.

Elsner, L. (1995). The generalized spectral-radius theorem: an analytic-geometric proof. *Linear Algebra and its Applications*, 220:151–159.

Finch, S. (2008). Private communication. Private communication.

Finch, S., Sebah, P., and Bai, Z.-Q. (2008). Odd entries in pascal's trinomial triangle.

Fornasini, E. and Valcher, M. E. (2010). Stabilizability of discrete-time positive switched systems. In *Proceedings of the 49th IEEE Conference on Decision and Control (CDC 2010)*, pages 432–437.

Goldwasser, J., Klostermeyer, W., Mays, M., and Trapp, G. (1999). The density of ones in pascal's rhombus. *Discrete Mathematics*, 204:231–236.

Golub, G. H. and Van Loan, C. F. (2013). *Matrix Computations*. Johns Hopkins University Press, Baltimore, MD.

Gripenberg, G. (1996). Computing the joint spectral radius. *Linear Algebra and its Applications*, 234:43–60.

Guglielmi, N., Manni, C., and Vitale, D. (2011). Convergence analysis of c^2 hermite interpolatory subdivision schemes by explicit joint spectral radius formulas. *Linear Algebra and its Applications*, 434:784–902.

Guglielmi, N. and Protasov, V. Y. (2013). Exact computation of joint spectral characteristics of linear operators. *Foundations of Computational Mathematics*, 13(1):37–97.

Guglielmi, N. and Protasov, V. Y. (2016). Invariant polytopes of linear operators with applications to regularity of wavelets and of subdivisions. *SIAM Journal on Matrix Analysis and Applications*, 37(1):18–52.

Guglielmi, N. and Protasov, V. Y. (2024). Computing the spectral gap of a family of matrices. *Mathematics of Computation*, 93(345):259–291.

Guglielmi, N., Wirth, F., and Zennaro, M. (2005). Complex polytope extremality results for families of matrices. *SIAM Journal on Matrix Analysis and Applications*, 27(3):721–743.

Guglielmi, N. and Zennaro, M. (2007). Balanced complex polytopes and related vector and matrix norms. *Journal of Convex Analysis*, 14:729–766.

Guglielmi, N. and Zennaro, M. (2008). An algorithm for finding extremal polytope norms of matrix families. *Linear Algebra and its Applications*, 428(10):2265–2282.

Guglielmi, N. and Zennaro, M. (2009). Finding extremal complex polytope norms for families of real matrices. *SIAM Journal on Matrix Analysis and Applications*, 31(2):602–620.

Gurvits, L. (1995). Stability of discrete linear inclusions. *Linear Algebra and its Applications*, 231:47–85.

Hare, K. G., Morris, I. D., Sidorov, N., and Theys, J. (2011). An explicit counterexample to the lagarias–wang finiteness conjecture. *Advances in Mathematics*, 226(6):4667–4701.

Hassan, M. F., Ivrissimitzis, I. P., Dodgson, N. A., and Sabin, M. A. (2002). An interpolating 4-point c^2 ternary stationary subdivision scheme. *Computer Aided Geometric Design*, 19(1):1–18.

Hechler, J., Mößner, B., and Reif, U. (2009). c^1-continuity of the generalized four-point scheme. *Linear Algebra and its Applications*, 430(11–12):3019–3029.

Horn, R. A. and Johnson, C. R. (1985). *Matrix Analysis*. Cambridge University Press, Cambridge.

Jungers, R. M. (2009). *The Joint Spectral Radius: Theory and Applications*, volume 385 of *Lecture Notes in Control and Information Sciences*. Springer, Berlin.

Jungers, R. M. and Protasov, V. Y. (2009). Counterexamples to the complex polytope extremality conjecture. *SIAM Journal on Matrix Analysis and Applications*, 31(2):404–409. Preprint: arXiv:0802.2654.

Jungers, R. M., Protasov, V. Y., and Blondel, V. D. (2008). Efficient algorithms for deciding the type of growth of products of integer matrices. *Linear Algebra and its Applications*, 428(10):2296–2312.

Jungers, R. M., Protasov, V. Y., and Blondel, V. D. (2009). Overlap-free words and spectra of matrices. *Theoretical Computer Science*, 410(38–40):3670–3684.

Kozyakin, V. (2005). A dynamical systems construction of a counterexample to the finiteness conjecture. In *Proceedings of the 44th IEEE Conference on Decision and Control and ECC 2005*, pages 2338–2343.

Kozyakin, V. S. (1990). Algebraic unsolvability of problem of absolute stability of desynchronized systems. *Automation and Remote Control*, 51(6):754–759.

Kozyakin, V. S. (2009). On the computational aspects of the theory of joint spectral radius. *Doklady Mathematics*, 80(1):487–491.

Lagarias, J. C. and Wang, Y. (1995). The finiteness conjecture for the generalized spectral radius of a set of matrices. *Linear Algebra and its Applications*, 214.

Maesumi, M. (1996). An efficient lower bound for the generalized spectral radius. *Linear Algebra and its Applications*, 240:1–7.

Maesumi, M. (2000). Joint spectral radius and hölder regularity of wavelets. *Computers & Mathematics with Applications*, 40(1):145–155.

Maesumi, M. (2008). Optimal norms and the computation of joint spectral radius of matrices. *Linear Algebra and its Applications*, 428(10):2324–2338.

Magaril-Il'yaev, G. G. and Tikhomirov, V. M. (2001). *Convex Analysis: Theory and Applications*, volume 222 of *Translations of Mathematical Monographs*. American Mathematical Society. Translated from Russian by Dmitry Chibisov.

Mason, O. and Shorten, R. N. (2007). Quadratic and copositive lyapunov functions and the stability of positive switched linear systems. In *Proceedings of the 2007 American Control Conference (ACC 2007)*, pages 657–662, New York City, NY, USA. IEEE. Marriott Marquis Hotel at Times Square, July 11–13, 2007.

Mejstrik, T. (2020). Improved invariant polytope algorithm and applications. *ACM Transactions on Mathematical Software*, 46(3):29:1–29:26.

Mejstrik, T. and Reif, U. (2025). A hybrid approach to joint spectral radius computation. *Linear Algebra and its Applications*.

Mengi, E. (2006). *Measures for Robust Stability and Controllability*. PhD thesis, New York University.

Möller, C. (2015). *A New Strategy for Exact Determination of the Joint Spectral Radius*. PhD thesis, TU Darmstadt.

Möller, C. and Reif, U. (2014). A tree-based approach to joint spectral radius determination. *Linear Algebra and its Applications*, 463:154–170.

Parrilo, P. A. and Jadbabaie, A. (2008). Approximation of the joint spectral radius using sum of squares. In *Linear Algebra and its Applications*, volume 428, pages 2385–2402.

Protasov, V. Y. (1996). The joint spectral radius and invariant sets of linear operators. *Fundamentalnaya i Prikladnaya Matematika*, 2(1):205–231.

Protasov, V. Y. (1997). The generalized spectral radius. a geometric approach. *Izvestiya: Mathematics*, 61(5):995–1030.

Protasov, V. Y. (2000). Asymptotic behaviour of the partition function. *Sbornik: Mathematics*, 191(3–4):381–414.

Protasov, V. Y. (2004). On the regularity of de rham curves. *Izvestiya: Mathematics*, 68(3):27–68.

Protasov, V. Y. (2010). Invariant functionals for random matrices. *Functional Analysis and Its Applications*, 44(3):230–233.

Protasov, V. Y., Jungers, R. M., and Blondel, V. D. (2010). Joint spectral characteristics of matrices: a conic programming approach. *SIAM Journal on Matrix Analysis and Applications*, 31(4):2146–2162.

Reznick, B. (1990). Some binary partition functions. In *Analytic Number Theory: Proceedings of a Conference in Honor of P.T. Bateman*, pages 451–477. Birkhäuser, Boston.

Rota, G.-C. and Strang, G. (1960). A note on the joint spectral radius. *Indagationes Mathematicae*, 22(4):379–381.

Rudin, W. (1976). *Principles of Mathematical Analysis*. McGraw-Hill, New York, 3 edition.

Stewart, G. W. and Sun, J. G. (1990). *Matrix Perturbation Theory*. Academic Press, New York.

Strang, G. (2001). The joint spectral radius (commentary). In *Collected Works of Gian-Carlo Rota*. Birkhäuser, Basel. Commentary on Rota–Strang (1960).

Tsitsiklis, J. N. and Blondel, V. D. (1997). The lyapunov exponent and joint spectral radius of pairs of matrices are hard—when not impossible—to compute and to approximate. *Mathematics of Control, Signals and Systems*, 10(1):31–40.

Vagnoni, C. and Zennaro, M. (2009). The analysis and the representation of balanced complex polytopes in 2D. *Foundations of Computational Mathematics*, 9(3):259–294.

Vandergraft, J. S. (1968). Spectral properties of matrices which have invariant cones. *SIAM Journal on Applied Mathematics*, 16:1208–1222.

Villemoes, L. (1994). Wavelet analysis of refinement equations. *SIAM Journal on Mathematical Analysis*, 25(5):1433–1460.

Zheng, H., Zhao, H., Ye, Z., and Zhou, M. (2007). Differentiability of a 4-point ternary subdivision scheme and its application. *IAENG International Journal of Applied Mathematics*, 36(1):19–24.